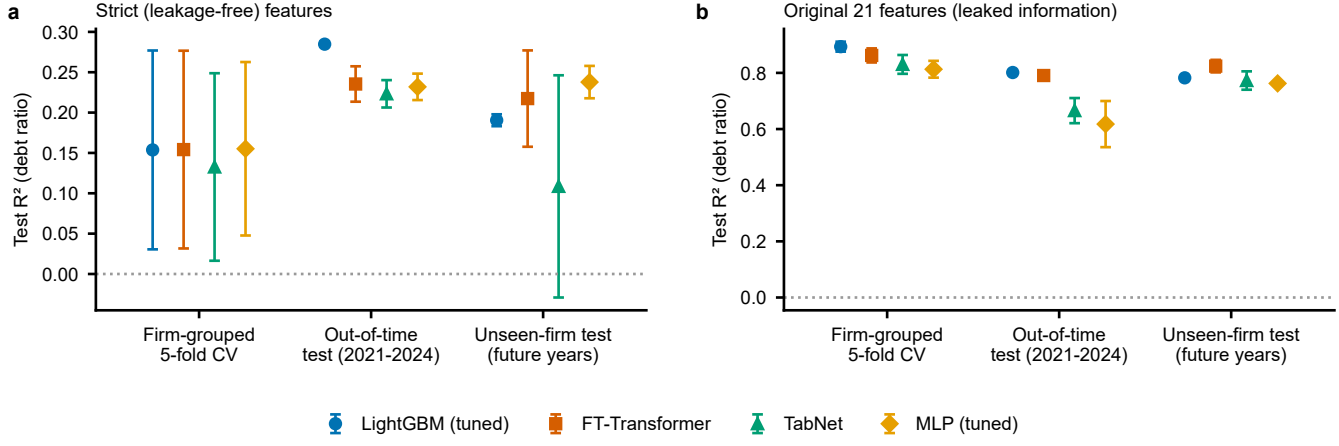
a
Strict (leakage-free) features
Test R² (debt ratio)
0.30
0.25
0.20
0.15
0.10
0.05
0.00
Firm-grouped 5-fold CV
Out-of-time test (2021-2024)
Unseen-firm test (future years)
b
Original 21 features (leaked information)
Test R² (debt ratio)
0.8
0.6
0.4
0.2
0.0
Firm-grouped 5-fold CV
Out-of-time test (2021-2024)
Unseen-firm test (future years)
LightGBM (tuned)
FT-Transformer
TabNet
MLP (tuned)